

Disproof of TLMC Conjecture 00000000589

The Frobenius number of $\langle n, n+1, n^2-n-1 \rangle$ is exactly $n^2 - 2n - 1$

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Abstract

Conjecture 00000000589 asserts that for every three-generated numerical semigroup $S = \langle a, b, c \rangle$ the Frobenius number satisfies $g(S) \leq \sqrt{3} (abc)^{1/3} (1 + o(1))$, with the constant $\sqrt{3}$ optimal. We disprove both claims with the family $S_n = \langle n, n+1, n^2-n-1 \rangle$: for every $n \geq 3$, $g(S_n) = n^2 - 2n - 1$ exactly, so $g(S_n)/(\sqrt{3} (abc)^{1/3}) \sim n^{2/3}/\sqrt{3} \rightarrow \infty$. The bound is first violated at $n = 6$ (ratio 1.2434) and fails by an unbounded multiplicative factor, so no constant can replace $\sqrt{3}$. All claims are verified in Lean 4 with zero axioms and zero `sorry`.

1 The conjecture

Conjecture 1 (00000000589). *Let $g(S)$ denote the Frobenius number of a numerical semigroup S . For every three-generated semigroup $S = \langle a, b, c \rangle$,*

$$g(S) \leq \sqrt{3} (abc)^{1/3} (1 + o(1)),$$

and the constant $\sqrt{3}$ is optimal (Erdős–Graham type).

2 The attack family

For $n \geq 3$ let

$$S_n = \langle n, n+1, n^2-n-1 \rangle, \quad N = n^2 - 2n - 1, \quad c_0 = n^2 - n - 1.$$

Note that $\gcd(n, n+1) = 1$, hence S_n is cofinite and $g(S_n)$ is defined. Also $c_0 = g(\langle n, n+1 \rangle)$: the third generator is precisely the Frobenius number of the first two.

Theorem 1. *For every $n \geq 3$,*

$$g(S_n) = N = n^2 - 2n - 1.$$

Proof. N is not representable. Suppose $N = xn + y(n+1) + zc_0$ with $x, y, z \geq 0$. If $z \geq 1$, the right side is at least $c_0 = n^2 - n - 1 > N = n^2 - 2n - 1$ (because $n > 0$), impossible. So $z = 0$ and, writing $u = x + y$,

$$N = xn + y(n+1) = un + y.$$

Reducing mod n and using $N = (n-3)n + (n-1)$ (valid for $n \geq 3$) gives $y \equiv n-1 \pmod{n}$, hence $y \geq n-1$. Since $u \geq y$,

$$un = N - y \leq N - (n-1) = (n-3)n,$$

so $u \leq n-3 < n-1 \leq y \leq u$, a contradiction.

Everything larger is representable. We show every $m \geq N+1 = n^2 - 2n$ lies in S_n , by strong induction on m .

- *Window.* If $m = n^2 - 2n + i$ with $0 \leq i \leq n - 2$, then $m = (n - 2 - i)n + i(n + 1)$, a combination of n and $n + 1$.
- *Generator point.* If $m = n^2 - n - 1 = c_0$, use the third generator.
- *Above.* If $m \geq n^2 - n$, then $m - n \geq n^2 - 2n$; by induction $m - n$ is representable and one extra copy of n gives m .

□

3 The ratio diverges

Corollary 1. *With $a = n$, $b = n + 1$, $c = c_0$,*

$$\frac{g(S_n)}{\sqrt{3}(abc)^{1/3}} = \frac{n^2 - 2n - 1}{\sqrt{3}(n(n + 1)(n^2 - n - 1))^{1/3}} \sim \frac{n^{2/3}}{\sqrt{3}} \rightarrow \infty.$$

Proof. Since $n^2 - 2n - 1 \sim n^2$ and $n(n + 1)(n^2 - n - 1) \sim n^4$, the quotient is asymptotic to $n^{2/3}/\sqrt{3}$. □

Corollary 2. *Conjecture 00000000589 is false. The inequality $g(S) \leq \sqrt{3}(abc)^{1/3}$ fails for every $n \geq 6$ in the family, and the ratio is unbounded, so the constant $\sqrt{3}$ is not merely suboptimal: no finite constant works for three-generated semigroups in this generality.*

Numerical check

n	$g(S_n)$	$\sqrt{3}(abc)^{1/3}$	ratio
3	2	6.782	0.295
4	7	10.457	0.669
5	14	14.361	0.975
6	23	18.497	1.243
8	47	32.017	1.468
12	119	72.315	1.646
20	359	93.869	3.824
100	9799	803.864	12.191

The exact values 14, 47, 119, 359 for $n = 5, 8, 12, 20$ (and all $3 \leq n \leq 60$) were confirmed by independent brute-force enumeration (`reproduce.py`).

4 Root-free certificates and Lean verification

The inequality $g > \sqrt{3}(abc)^{1/3}$ for positive g, abc is equivalent to $g^6 > 27(abc)^2$; since $3\sqrt{3} < 5$, the weaker rational certificate $g^3 > 5abc$ suffices. The Lean development checks, e.g.,

$$5 \cdot 1218 = 6090 < 12167 = 23^3 \quad (n = 6), \quad 5 \cdot 159180 = 795900 < 46268279 = 359^3 \quad (n = 20),$$

$$5 \cdot 99979900 = 499899500 < 940903909399 = 9799^3 \quad (n = 100).$$

The Lean 4 sources (toolchain `leanprover/lean4:v4.33.1`, no Mathlib) prove Theorem 1 for *arbitrary* $n \geq 3$ as `frob_family`, instantiate it at $n = 6, 20, 100$, and check the certificates by `decide`. The audit `Check.lean` shows that all 31 declarations (including the hand-rolled truncated-subtraction toolkit `my_*`) satisfy

...does not depend on any axioms

i.e. no `sorryAx`, no `Classical.choice`, no `Quot.sound`, no `propext`. Reproduce with `lake build && lake env lean Check.lean` inside `lean4/`; the numerics with `python3 reproduce.py`.

5 Boundary remarks

- $n = 2$: $n^2 - n - 1 = 1$, so S_2 contains 1; the Frobenius number is undefined and the family starts at $n = 3$.
- $n = 3, 4, 5$ still satisfy the conjectured bound; $n = 6$ is the first counterexample.
- The construction shows why the conjecture cannot be repaired by any constant: the two-generator obstruction $g(\langle a, b \rangle) \sim ab$ is carried into the three-generated world by taking the third generator $c = g(\langle a, b \rangle)$, which lowers the Frobenius number only marginally (here from $n^2 - n - 1$ to $n^2 - 2n - 1$).